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(See 6.3)

MILITARY SPECIFICATION

MAINTENANCE PRINTS FOR ELECTRONIC EQUIPMENT, INSTRUCTIONS FOR PREPARATION OF

1. SCOPE

1.1 This specification covers the preparation and format of maintenance prints which are to be supplied with the electronic equipment to facilitate maintenance and servicing. To aid the technician in trouble-shooting, certain information from the Technical Manual and from other sources is condensed to form maintenance prints which will enable the service technician to perform his task more efficiently.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-535 - Plastic Sheet (Sheeting and Film); Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.

MILITARY

MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-L-17192 - Lubrication Design, Lubricants, and Lubrication Information for Electronic Equipment; General Specification.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN STANDARDS ASSOCIATION (ASA)

Y32.16 - Electrical and Electronics Reference Designations.

(Application for copies should be addressed to the American Standards Association, Inc., 10 East 40th Street, New York, N.Y. 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

FSC 7650

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3. REQUIREMENTS

3.1 General.- To establish uniformity in the preparation, format, and general information content, maintenance prints shall be in accordance with the requirements specified herein and the individual prints shall consist of the following:

(a) Technical characteristics (see 3.2).
(b) Lubrication chart (see 3.3).
(c) Servicing-block diagram (see 3.4), and its associated material.
(d) External cabling diagram (see 3.5).
(e) Primary power breakdown schematics (see 3.6).
(f) (Radar) servo system block diagram (see 3.7).
(g) Major unit schematics (see 3.8).
(h) Photographs of equipment and alignment charts (see 3.9).

3.1.1 Unless otherwise specified herein, all paragraphs contained within this specification shall refer to both radar and radio equipment, and similar electronic equipment, as applicable.

3.2 Technical characteristics.- The following is a sample list of technical characteristics which are considered significant to maintenance personnel:

Radio

(a) Transmitter:
    (1) Frequency range.
    (2) Number of preset frequencies.
    (3) Type of frequency control.
    (4) Crystal type and data.
    (5) Frequency stability data.
    (6) Type(s) of emission and modulation capabilities.
    (7) Nominal unmodulated power output.
    (8) Maximum modulated power output.
    (9) Power requirement.
    (10) Unusual features.
    (11) Antenna-dummy load characteristics.
(b) Receiver:
    (1) Frequency range.
    (2) Type receiver (for example, superheterodyne).
    (3) Type of reception.
    (4) Sensitivity and signal to noise ratio.
    (5) Intermediate frequency.
    (6) Band width.
    (7) Receiver output power.
    (8) Audio frequency response.
    (9) Output impedance and input impedance.
    (10) Power requirements.
    (11) Unusual features.

Radar

(a) Antenna assembly:
    (1) Type (for example, parabolic).
    (2) Dimensions.
    (3) Beam width(s).
        a. Horizontal.
        b. Vertical.
    (4) Side lobe and back (90 through 270 degrees) radiation.
    (5) Gain.

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(6) Scan rate.
    a. Azimuth.
    b. Elevation.
(7) Type drive system.
(8) Unusual features.

(b) Transmitter/modulator (list both if separate units):
(1) Frequency range.
(2) Wave-length.
(3) Transmitter and duplexer tube types.
(4) Average power.
(5) Peak power.
(6) Pulse width(s).
(7) Pulse repetition rate(s).
(8) Power requirements for operation.
    a. Voltage.
    b. Frequency.
    c. Wattage.
    d. KVA.
    e. Power factor.
(9) R.f. isolator(s).
(10) Unusual features.

(c) Receiver:
(1) Local oscillator tube type.
(2) Noise figure (db) (inclusive of duplexer).
(3) Intermediate frequency.
(4) I.f. Bandwidth.
(5) Sensitivity.
(6) Unusual features.

(d) Signal comparator (moving target indicator):
(1) Type.
(2) Clutter cancellation ratios.
(3) Unusual features.

(e) Indicator(s)
(1) Type(s) and size(s).
(2) Range scale(s).
(3) Unusual features.

(f) System characteristics:
(1) Weight and size
    a. Transport.
    b. Operating.
(2) Input power requirements.
(3) Erection time and crew required.

3.3 Lubrication chart. -

3.3.1 Prints shall be supplied to describe the lubrication procedure of the entire equipment, and shall be in accordance with the requirements of this specification.

3.3.2 Unless otherwise specified in the contract or order, lubrication charts shall contain the following data and instructions:

(a) Illustration of major unit(s) or accessory to be lubricated shall have full identification, nomenclature, and type (if any).
(b) Individual lubrication points shall be identified by arrows or callouts. Where lubrication points are indistinguishable on the illustrations, or where many lubrication points are in a confined area, blow-up illustrations showing these points shall be included on the chart.

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(c) The following information shall be at the tail-end of the arrow and in the given order:
  (1) Letter(s) indicating lubrication interval (see 3.3.2(d)).
  (2) Short designation of lubricant (see applicable table of MIL-L-17192).
  (3) Name of part to be lubricated and number of lubrication points.
  (4) Detailed instruction for applying the lubricant, including disassembly. If extensive instructions for application of lubricant or disassembly of equipment are necessary, refer to a note at the bottom of the chart or to the applicable section of the technical manual. Specify how much lubricant to apply, how to apply the lubricant, the tool or device used to apply lubricant, and any precautions to be taken. Specify sizes of fill and drain plugs.

(d) A legend explaining the letter which indicate the lubrication intervals is required (for example):
  D - Daily
  W - Weekly
  M - Monthly
  SA - Semiannually
  A - Annually
  BE - Biennially (every two years)
  O - Only when overhauled
  X, Y, Z - As required (explain circumstances)

(e) A tabular listing of tools required (see table I) and general description (see table II), shall be used as follows: Tables I and II shall serve as illustrations and shall include the information necessary for the particular equipment:

Table I - (Sample) table of tools required.

Tools required for lubrication				
Description	Federal stock number	Standard Navy stock number	Signal Corps stock number	Standard Air Force stock number

Table II - (Sample) stock number table.

Lubricant							
Type of lubricant	Grade	Federal stock number	Standard Navy stock number	Signal Corps stock number	Standard Air Force stock number	For use in extreme emergency only	
						Below 50° F.	Above 50° F.

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3.3.3 When more than one command or agency is concerned, the stock numbers in tables I and II used by those agencies shall be given and shall be obtained from each agency respectively. Also, when available, Federal stock numbers shall be given.

3.4 Servicing-block diagram.-

3.4.1 The servicing-block diagram and its associated material shall enable Military personnel to isolate and localize a point of failure quickly.

3.4.2 Servicing-block diagram and its associated material shall be depicted as shown on figures 1 or 6, 2 and 3.

3.4.3 Insofar as possible, the sequences of blocks for a particular channel, for example, the video stages, shall be depicted from the left to the right of the sheet and in one line in order to minimize reversal of direction of flow.

3.4.4 The following shall apply for both radar and radio equipment, as applicable:

3.4.4.1 The transmitter-duplexer-receiver shall be shown along the top of the sheet, starting from the left and continuing to the right side of the print.

3.4.4.2 The servicing-block diagram shall include the modulating, video, synchronizing, sweep, (un)blanking, transmitting, IFF, receiving, duplexing, and waveguide-antenna circuits. In addition, if applicable, monitor scope, r.f. monitor, oscillation indicator meter, echo box, and similar units shall be briefly depicted. Power supplies, regulated rectifiers, synchro system and similar units shall not be shown. However, power suppliers and other units closely associated with modulator system shall be shown.

3.4.4.3 Video, sweep, (un)blanking, and marker channels shall be sufficiently identified as such.

3.4.4.4 Differentiating and integrating circuits shall be shown to include the resistor and capacitor (or inductance) combination involved together with the component symbol designation. The abbreviations DIFF. or INT. shall be placed near the component. In addition, the waveforms at the input and output of the differentiator and integrator shall be included (see 3.11).

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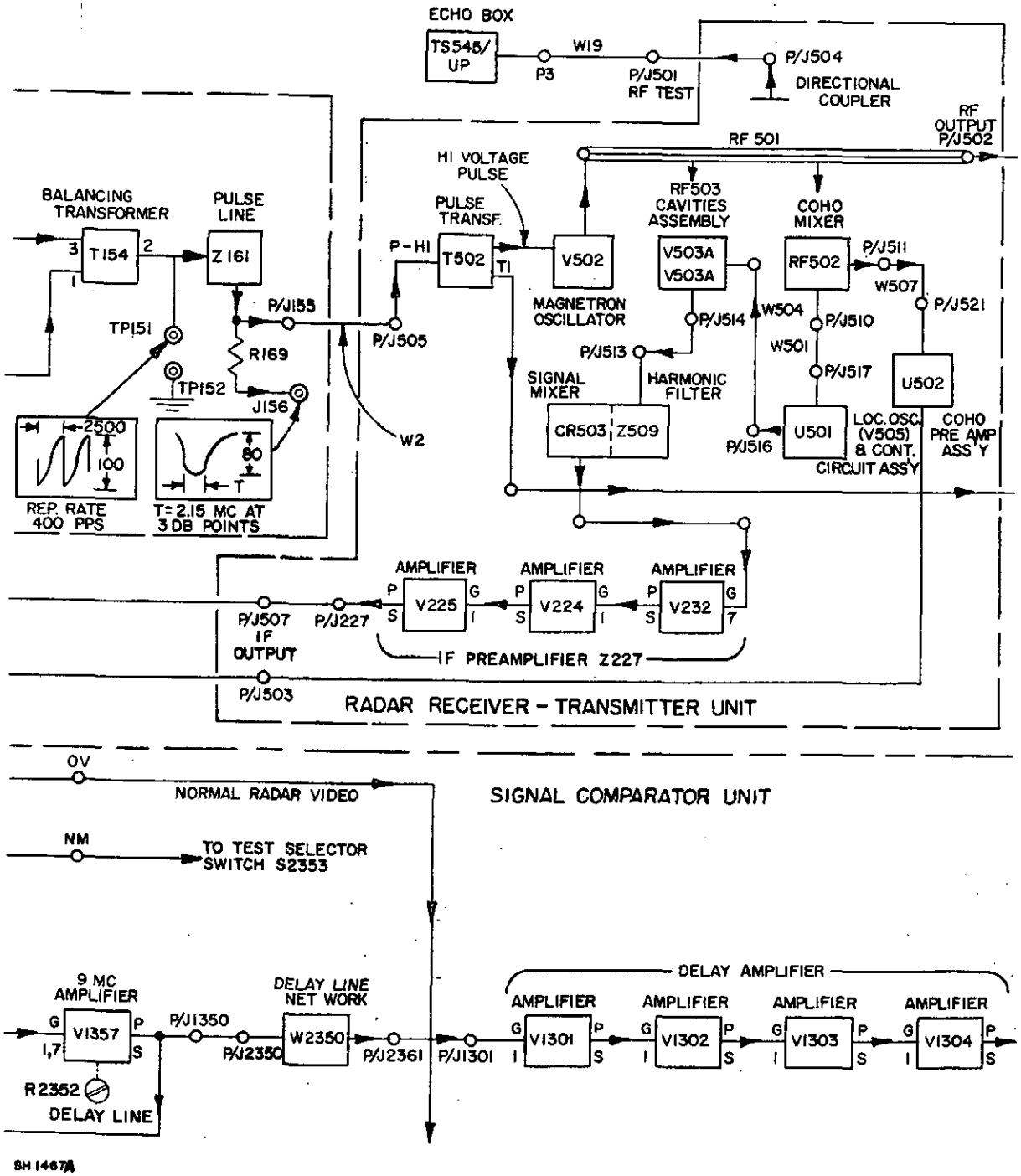


Figure 1 - Servicing-block diagrams.

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Voltage Measurement.

Symbol number	Tube number	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8	Pin 9
V-102	12AU7	300	0	16.5	6.3 vac	6.3 vac	175	-0.2	4	0
V-103	12AX7	232	86	2.5	6.3 vac	6.3 vac	275	2.5	20	0
V-104	6AU6	-47	0	6.3 vac	0	2.2	250	0	--	--
V-210	6AL5	73	58	6.3 vac	0	71	N.C.	---	--	--
V-211	1Z2	N.C.	**V	N.C.	N.C.	N.C.	**V	N.C.	--	--
V-301	6L6	0	6.3 vac	460	300	-46	N.C.	0	4.7	--
V-401	2C40	N.C.	6.3	0	N.C.	0	N.C.	0	0	--

Resistance measurement.

Symbol number	Tube number	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8	Pin 9
V-102	12AU7	20K	650K	1.4K	0	0	20K	1.2K	1.4K	0
V-103	12AX7	25K	800K	1.4K	0	0	10K	14K	20K	0
V-104	6AU6	2.5 meg.	0	0	0	5 meg.	60K	0	---	--
V-210	6AL5	12K	2.5 meg.	0	0	71	N.C.	---	---	--
V-211	1Z2	N.C.	8 meg.	N.C.	N.C.	N.C.	8 meg.	N.C.	---	--
V-301	6L6	0	0	1NF	9K	300K	N.C.	0	350	--
V-401	2C40	N.C.	0	0	N.C.	0	N.C.	0	0	--

Notes:
N.C. - No connection.
** - High voltage (do not measure).

Figure 2 - Voltage and resistance measuring chart.

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Fuse locations				
Symbol	Protects	AMP	Volts	Number
F401	Service outlet J408	1.6	250	FNM-1
F402	System power control relay	1.6	250	FNM-1
F403	Magnetron cooling blower motor B601 IF Preamplifier filament transformer T503 Magnetron filament transformer T604	4	250	FNM-4
F404	Indicator blower motor B601 Indicator filament transformer T605	6.25	250	FNM-6-1/4
F405	Signal Comparator blower motor B2350 Signal Comparator filament transformer T2350	1.6	250	FNM-1
F406	Power Supply filament-type transformer T403 Antenna synchro motor B752 Antenna blower motor B755 Grid voltage transformer T701, for V701 and V702 Antenna filament transformer T702	6.25	250	FNM-6-1/4
F407	Plate supply transformer T704, for V701 and V702	0.3	250	FNM-0.3
F408	Plate supply transformer T704, for V701 and V702	15	250	BAN-15
F409	Circulation blower motor B401 Power supply filament transformer T401 Power supply plate power transformer T402 Keep-alive voltage transformer T501 Power transformer T604	6.25	250	FNM-6-1/4
F410	High-voltage plate transformer T151 Trigger chassis filament transformer T101 Modulator rectifier filament transformer T152 Modulator thyratron filament transformer T153	6.25	250	FNM-6-1/4
F411	High-voltage plate supply transformer T151	30	250	BAN-30
F412	System power	60	250	FRN-60
F413	System power	60	250	FRN-60
F414	Heater HR401	30	250	NON-30
F415	Heater HR401	30	250	NON-30
F416	Service outlet, standby	6.25	250	FNM-1

Test point locations	
Modulator	
TP101-TP105	Top trigger subchassis, lower front of modulator chassis
TP151 and TP152	Lower right side of chassis
J153, J154, and J156	Lower right of front panel
Indicator	
TP601 and TP602	Lower front of rear subchassis
TP603	Right front of rear subchassis
TP604	Lower front of rear subchassis
TP605-TP608	Top front of rear subchassis
TP609 and TP610	Left front of rear subchassis
Signal comparator	
TP301	Lower left, front panel
Antenna base	
TP701 and TP702	Top Servo Amplifier chassis

Figure 3 - Test point and fuse location chart.

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3.4.5 The functions and symbol designation of the transformer, delay and pulse forming lines shall be depicted as shown on figure 1.

3.4.6 The grid input and plate output of tube blocks shall be at the left and right side respectively and shall be designated by element abbreviation and pin numbers. When possible, these designations shall be separated by the line connecting the blocks - the element abbreviation being above the line and the pin number below the line. When the output is taken from the cathode, it shall be shown as leaving the bottom or the top instead of the right side of the block (see figures 1, 4 and 5 and 3.4.6.3).

3.4.6.1 Each tube shall be shown as a separate block except for dual tubes, such as type 6SN7WGTA, which shall be shown as one long block (see figure 1 or 5).

3.4.6.2 When the two halves of a 6SN7WGTA or similar tubes do not appear adjacent to one another but are located in different sections of the circuit, the block shown on figure 4 shall be used, for each tube half, in lieu of that shown on figure 5.

3.4.6.3 The tube type and circuit symbol shall be shown within the block; the function just above the block, as shown on figures 4 and 5 (also see figure 1).

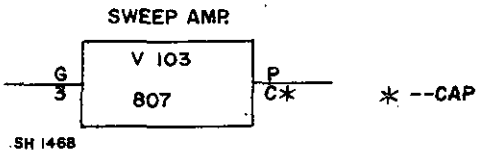


Figure 4 - Tube block diagram.

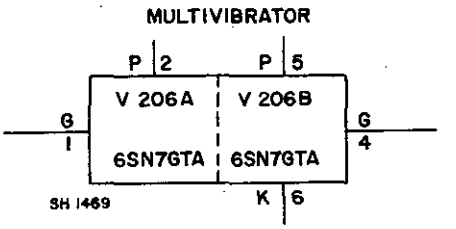


Figure 5 - Tube block diagram.

3.4.6.4 The blocks used on the servicing-block diagram shall have the following proportions with the longer dimensions placed horizontally:

(a) Blocks representing single tubes (or half of dual tubes) = 5 x 4.
(b) Blocks representing dual tubes = 10 x 4.
(c) Blocks representing other parts (such as transformers, delay and pulse forming networks necessary to depict all functions = 4 x 4.

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3.4.7 For radio equipments, the layout of the servicing-block diagram shall be similar to the one shown on figure 6. The blocks used shall be in accordance with 3.4.6.4.

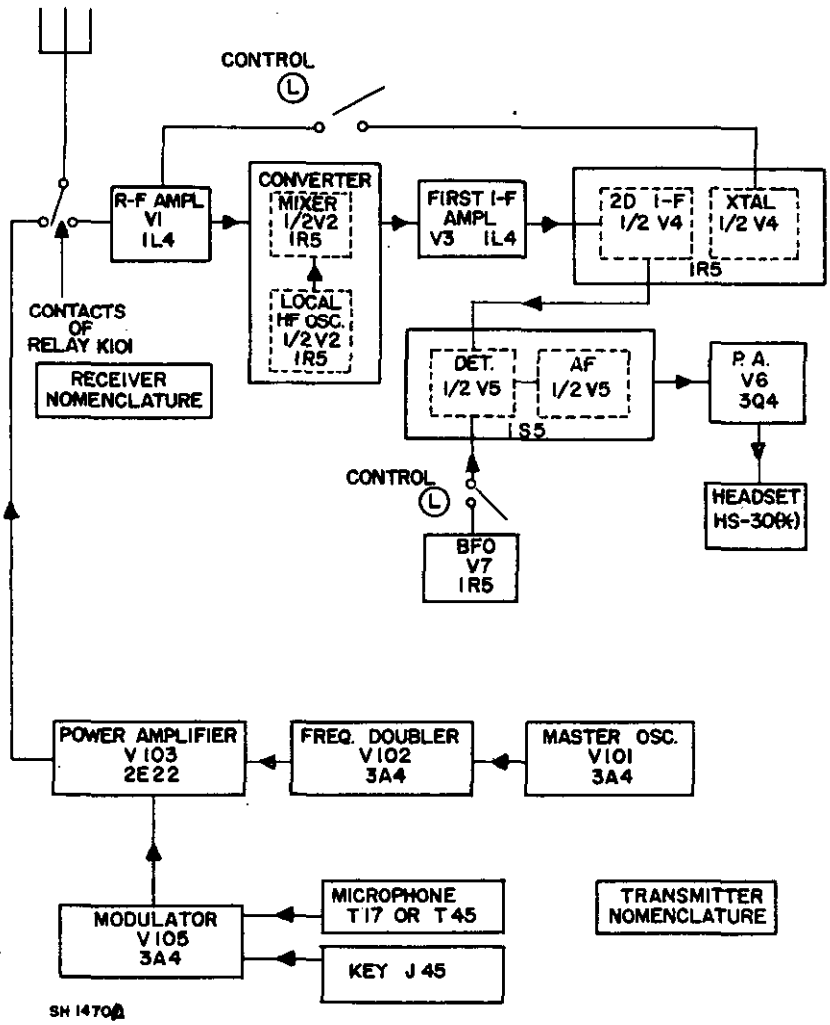


Figure 6 - Servicing-block diagram for radio equipment.

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3.4.8 The title of the servicing-block diagram shall contain the following information:

RADAR (RADIO) SET

(NOMENCLATURE OF EQUIPMENT)

SERVICING-BLOCK DIAGRAM

3.4.9 The material associated with the servicing-block diagram shall consist of test point and fuse location tables, voltage measurement tabulations, and resistance measurement tabulations and shall be printed on one page if overcrowding does not result. The title of the page(s) shall be as follows:

TEST INFORMATION

FOR USE WITH

(NOMENCLATURE OF EQUIPMENT)

SERVICING BLOCK DIAGRAM

3.4.10 Voltage measurement tabulation.- A voltage measurement chart shall give the plate, grid, screen-grid, cathode, and other similar d.c. voltages present at the terminals of all the tubes (including the cathode ray tubes) used throughout the entire equipment (see figure 2). Heater voltages shall also be included.

3.4.10.1 All voltage measurements that are taken with the voltmeter shall be made on the lowest scale possible so that the percent of error shall be at a minimum.

3.4.10.2 The type of meter (electronic voltmeter or nonelectric multimeter) and the resistance in terms of ohms per volt of the voltmeter used shall be indicated. In addition, other pertinent information as to operating conditions which may influence the magnitude of these voltages shall be given.

3.4.11 Resistance measurement tabulation.- A resistance measurement chart shall be furnished, listing resistance measurements taken at each tube pin for all tubes (including cathode ray tubes) used throughout the entire equipment; tubes shall be identified by both type and symbol (see figure 2).

3.4.11.1 All resistance measurements that are taken with the ohmmeter shall be made on the lowest scale possible so that the percent of error shall be at a minimum.

3.4.11.2 Pertinent information regarding operating conditions which may influence the magnitude of these resistances shall be included.

3.4.12 Fuse information chart.- The physical location of all fuses used throughout the entire equipment shall be included (see figure 3).

3.4.12.1 Information as to the symbol number, current rating, location, function, and voltage to ground (at a specific test point) also shall be included.

3.4.13 Wave forms.- Photographs of the actual wave form taken at the input and output terminals of each tube and at other points if deemed necessary, shall be depicted with an arrowhead pointing to the terminal or point at which the wave is obtained (see figure 1).

3.4.13.1 Photographs shall be of such clarity as to enable the technician to rapidly determine whether or not a circuit is functioning properly (see figure 1).

3.4.13.2 Sweep circuit and (un)blanking wave forms shall be shown for only one setting of the range control unless the wave forms produced at various ranges are appreciably dissimilar in shape and voltage, or both.

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3.4.13.3 In those instances when the waveform is of the same shape and amplitude at the plate of a tube and at the grid of a succeeding tube, only one waveform shall be shown between tube blocks with one distinct arrow leading from the waveform and terminating at the test point.

3.4.13.4 The following information shall be shown adjacent to each waveform (see figure 1).

(a) Sweep frequency (SF) of the test oscilloscope.
(b) Duration in microseconds (for example: T = 20 μsec.).
(c) Peak voltage (for example: V = 95 volts)
(d) Range at which waveform is taken (for example: R = 10 miles).

3.4.13.5 *Waveforms that are shown on the servicing-block diagram shall be those observed on one of* the following model oscilloscopes which are listed in the order of preference:

(a) Test oscilloscope supplied with the equipment.
(b) AN/USM-32 (Marine Corps).
(c) OS-8B/U or OS-8C/U Navy.

3.4.13.5.1 If the oscilloscopes specified in 3.4.13.5 are not available, the contractor shall obtain written approval from the command or agency concerned as to the make and model number of the oscilloscope that is required or to be used for the particular application.

3.4.13.6 A sweep of such frequency as to show clearly the true waveform shall be employed. However, the number of distinct frequencies shall be minimized insofar as possible to permit the technician to use the same sweep frequency for the greatest number of observations. Whenever possible, the vertical amplifier of the test oscilloscope shall be turned off when observations are made. Waveforms shall not be shown inverted. If the test oscilloscope inverts the waveform a note by the waveform shall state this.

3.4.13.7 Duration of only the significant part(s) of the waveform shown at a point in the circuit shall be designated by dimension lines and the letters "T", T_1", on the waveform reproduction itself. A note by the waveform shall indicate the duration in microseconds represented by the letters "T", "T_1", for example T = 20 μsec. The word "significant" is defined as follows: Those instances in which the operation of the circuit itself or succeeding circuit(s) is based upon specific portion(s) of the waveform.

3.5 External cabling diagram.-

3.5.1 The external cabling diagram shall show all self-contained units, for example, power supply, signal comparator, antenna base, receiver, and transmitters, of the equipment as blocks with interconnecting cables and cables leading to other units (see figure 7).

3.5.2 Each cable (including coaxial cables) shall be shown as one line (except where fanned out at terminals and plugs) regardless of the actual number of conductors involved. Cables shall be plainly identified as to function (if other than an electrical power cable), designation, and type number (if any). If the total number of conductors in a cable is not utilized, a note shall be placed by the cable indicating the number of active conductors (see figure 7).

3.5.3 The following shall be included inside each self-contained unit (see figure 7):

(a) Plugs and jacks.
(b) *Terminal strips and terminals.*
(c) Plug and jack numbers and pin numbers.
(d) Terminal strip numbers and terminal numbers.
(e) Name, nomenclature and type numbers, if any, of the unit.

3.6 Primary power breakdown schematic.-

3.6.1 The primary power breakdown diagram is an "across-the-line" breakdown showing all parts connected across or in series with the a.c. or d.c. bus supplying power to the electronic equipment (for example: radar equipment - see figure 8).

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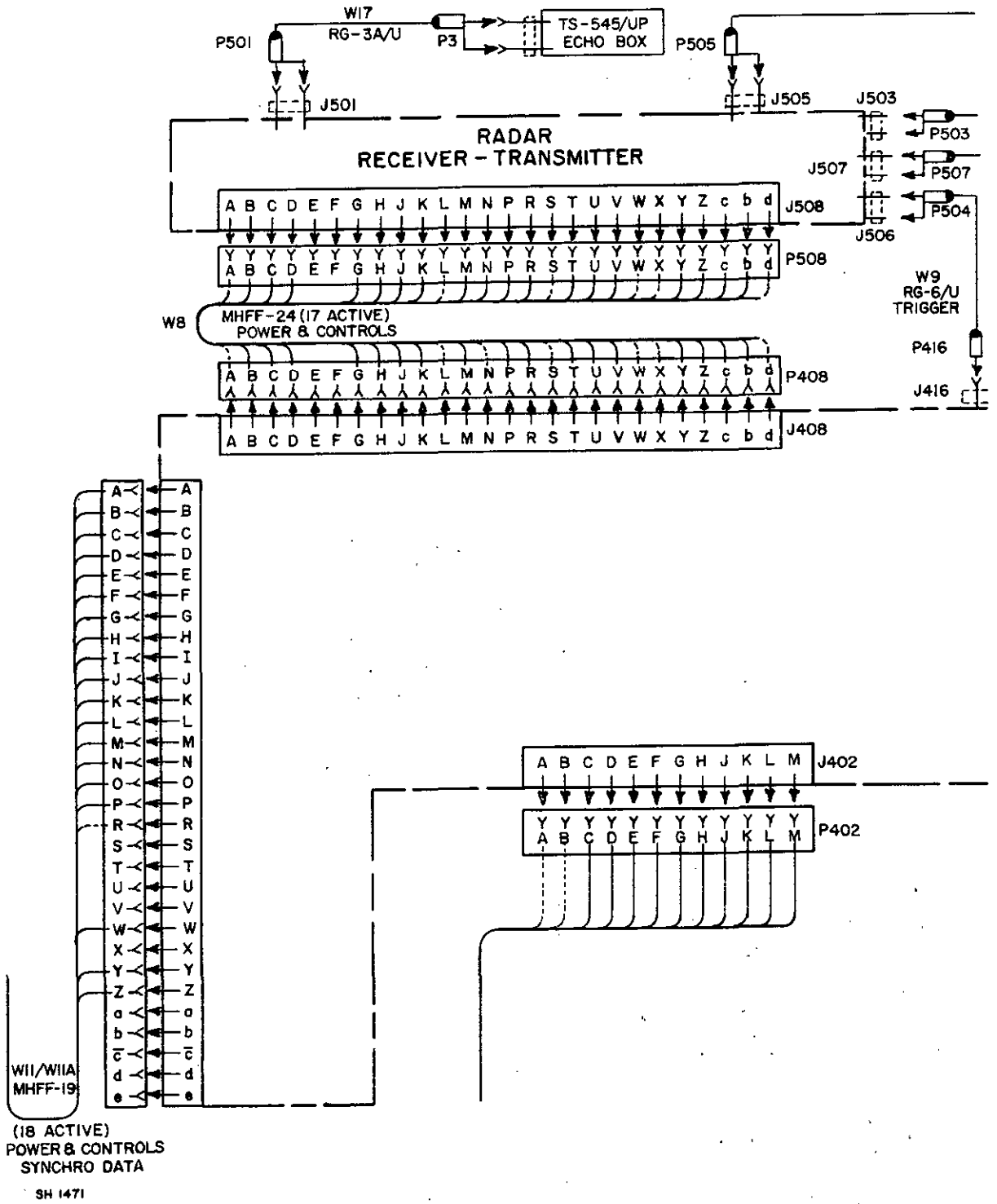


Figure 7 - External cabling diagram.

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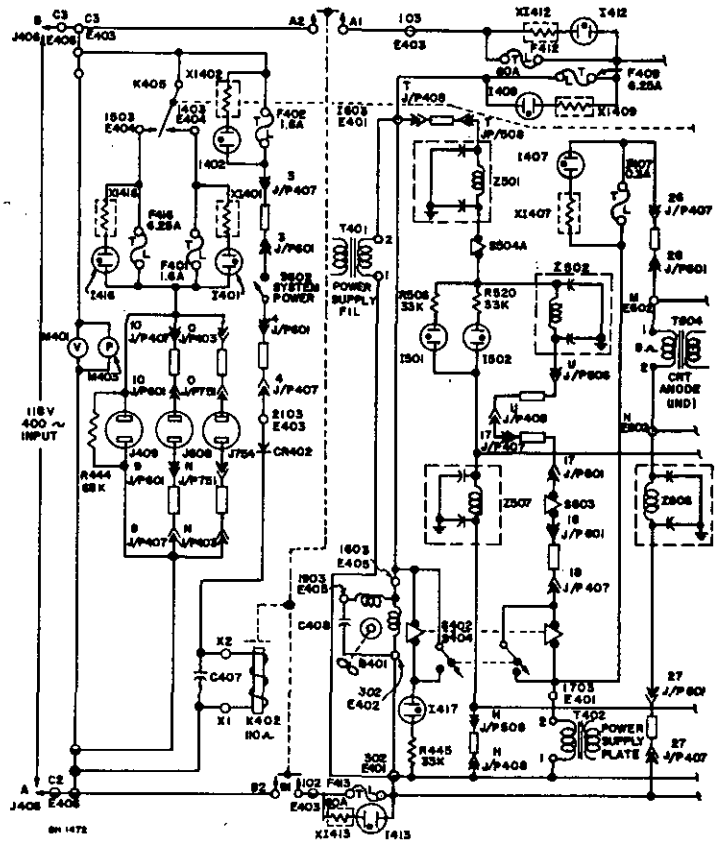


Figure 8 - Power distribution and d.c. control circuit diagram.

3.6.2 Among others, the following shall be shown schematically on the diagram (for example):

(a) All fuses.
(b) Voltmeters and ammeters.
(c) Relay solenoids.
(e) Primary windings of transformers.
(e) Interlock switches.
(f) Pilot lamps.
(g) Blower motors.
(h) Excitation to synchro units.
(i) Main frame space heaters.
(j) Antenna pedestal space heaters.
(k) Main frame line switch.
(l) Overload reset switches.
(m) Antenna thermostats.
(n) Interference filters.
(o) Radiation switch.
(p) Auto transformers.
(q) Voltage regulator transformers.

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(r) "Standby" - "ON" switch.
(s) Interlock switch bypasses.
(t) Motor dynamo amplifiers.
(u) All terminals on terminal-boards and plug and jack pin to which primary power lines and parts are attached.

3.6.3 Relay contacts shall be shown adjacent to associated solenoids only when such association does not detract from the clarity of the diagram.

3.6.4 To aid in tracing the continuity or primary power circuits, distinctive marking shall be placed at terminal points to indicate equipotential points, for example, a common or grounded side of a two conductor circuit shall be indicated by a small circle with the bottom half darkened (◒); the switched or high side of the same circuit by a small circle (◒); and regulated circuit by a small circle with the vertical half of the circle darkened (◑) (see figure 8). A legend shall be printed on the sheet to explain the meaning of these symbols.

3.6.5 The function of all parts shall be listed in a table in an alphabetical-numerical sequence near the diagram itself by symbol in the manner indicated hereinafter. Under the function column, the ratings of fuses shall be given:

Symbol	Function
B-101	Main blower motor
F-108	6A Renewable fuse link
F-703	3A glass fuse
T-103	Modulator H.V. plate

3.6.5.1 In addition to the requirement of 3.6.5, information shall be placed by the parts to indicate their functions, for example, by a voltmeter symbol, the word "Line"; by a transformer primary, the word "Mag. Fil".

3.6.6 All terminals shall be marked on the diagram to correspond to the marking on the actual equipment terminal. In those instances where no specific system is used which will immediately identify a terminal with relation to the unit of the equipment in which it is located, this information shall be listed (alphabetically or numerically by terminal number). For example:

Terminal	Unit
12	Trans
14	Mod. Gen.
D-1	Ant. Ped.

3.6.6.1 When a specific system of numbering is used throughout the equipment, for example, when all terminals in the modulation generator unit are in the 600 series group, or all terminals in the transmitter unit are in the 300 series group, the information given in 3.6.6 will not be required. Instead, the following information shall be given on the sheet on which the diagram is printed:

Series	Unit
100 to 200	PPI
200 to 300	Receiver

3.7 (Radar) servo system block diagram(s).-

3.7.1 All servo systems shall be illustrated by means of a block diagram (see figure 9). If functionally interconnected, they shall be portrayed on a single diagram; if not, separate diagrams shall be used.

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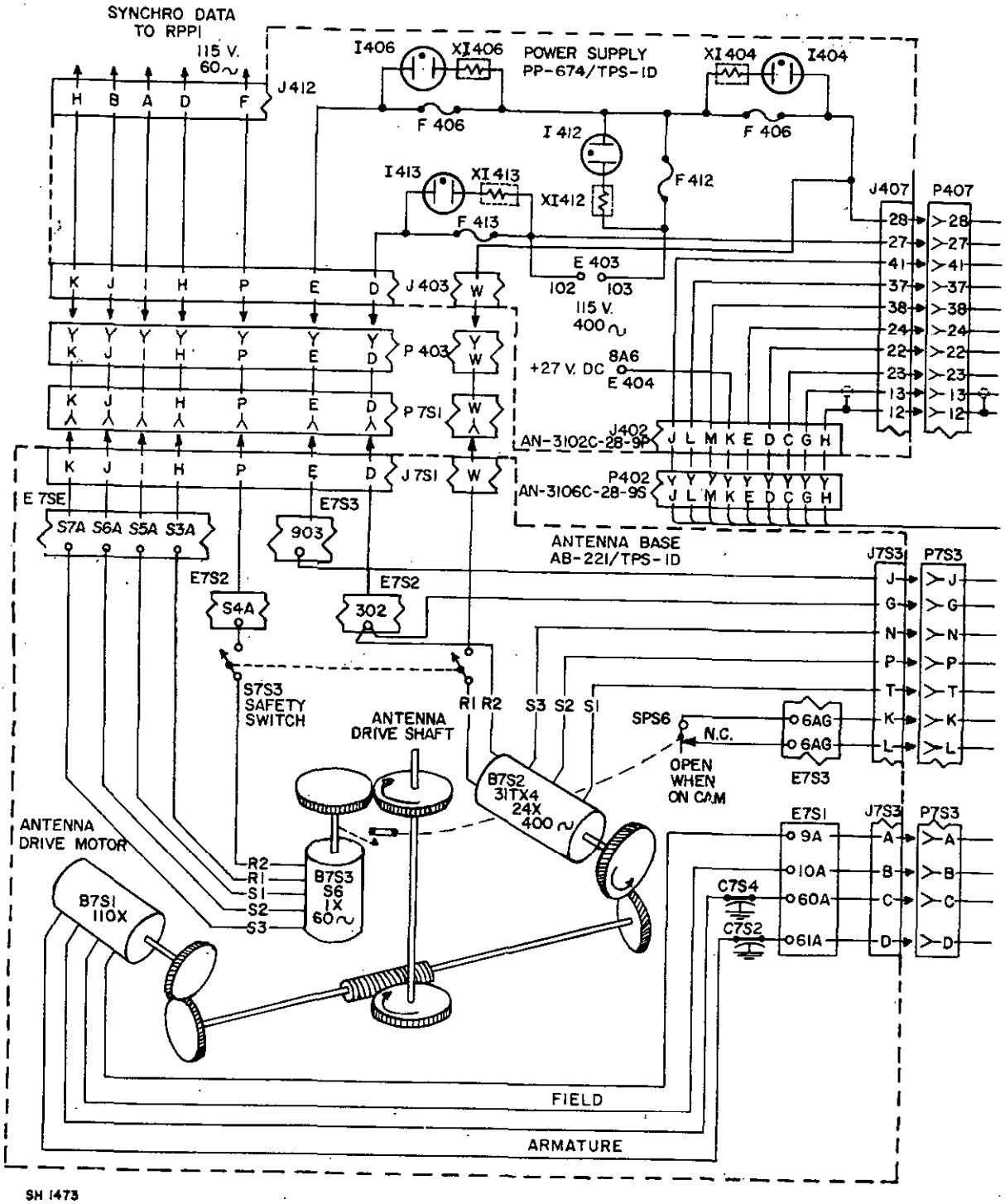


Figure 9 - Servo system diagram.

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3.7.2 Servo units, control amplifiers, antenna drive motors, motor dynamo-amplifier, auto transformers, relays, synchro capacitors, switches, terminal boxes, terminals (with numbers), terminal strips, and jacks and plugs, and all major parts associated with the servo control system shall be shown.

3.7.3 Servo units shall be shown as blocks (see figure 9) labeled with symbol number (for example, "B-702") type (for example, "5 CT synchro") and frequency (60 cycles). In addition, synchro units shall be shown geared to their load or their driving input. Synchro speed ratios and gear ratios shall also be given.

3.7.4 Subunits and terminal strips within major assembly or unit shall be enclosed by dashed lines (see figure 9). The enclosure shall be clearly labeled, for example: "Antenna Assembly"; "Control Amplifier"; "Range Indicator".

3.7.5 Control amplifiers and dynamo-amplifiers shall be shown as blocks. Synchros and motors shall be represented isometrically.

3.7.6 The magnitude and source of exciting voltage to servo units and antenna drive motors shall be shown legibly.

3.7.7 The output to remote bearing and range circuits shall be so designated.

3.8 Major unit schematics.-

3.8.1 A schematic diagram of each of the major units shall be furnished. If, for example, a major unit contains both the transmitter and receiver, the combined prints shall be shown on one page unless in so doing legibility is sacrificed.

3.8.2 Two schematics of smaller units may be placed on one sheet to conserve material and reduce the bulk of the material furnished if legibility is not sacrificed and crowding does not result.

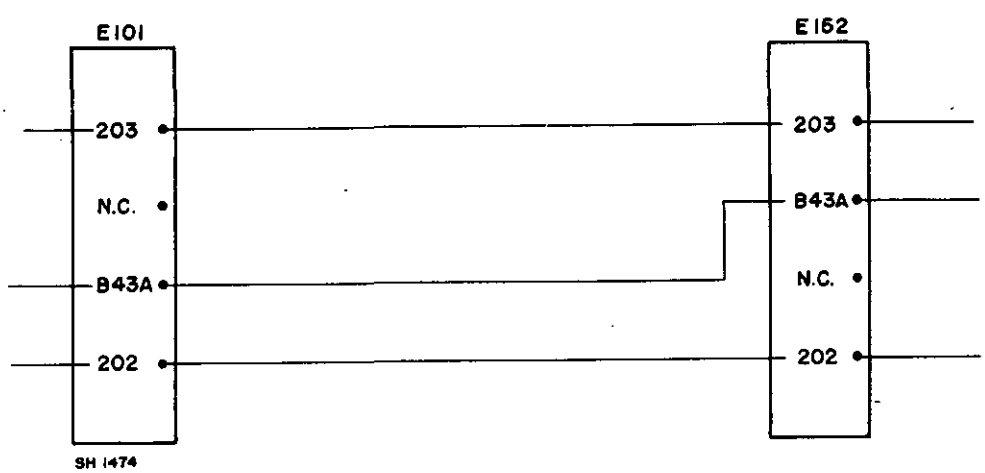


Figure 10 - Interconnection between terminal strips.

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Figure 11 - Interconnection between plugs and jacks.

3.8.3 Terminal strips, terminals, and plugs and jacks shall be shown legibly.

3.8.4 All cables shall be shown and identified as to type number, designation and function, for example, "video cable", "oscillator cable", and "mixer cable".

3.8.5 Where wires or cables connect from terminals within a unit to other units of equipment, the system shown on figure 10 shall be used to indicate the corresponding terminals and terminal strips of the unit external to the one shown.

3.8.5.1 Where interconnections are made with plugs and jacks, the system shown on figure 11 shall be used.

3.8.6 Controls shall be identified both as to function, symbol number, and electrical value.

3.8.7 Each schematic shall have a title box which will include the nomenclature and the title of the diagram.

3.8.8 Emphasis shall be placed on the layout of the schematic, size of parts depicted, and legibility of printing, for ease of reading and convenience in trouble-shooting.

3.9 Photographs of the equipment and alignment charts.-

3.9.1 All replacement parts contained in each major equipment shall be shown by photographic illustrations so taken as to offer the best view of the part(s).

3.9.2 Designation arrows or callouts (with symbol number of the part accompanying) shall be used to show the location of each replacement part in the manner shown on figure 12.

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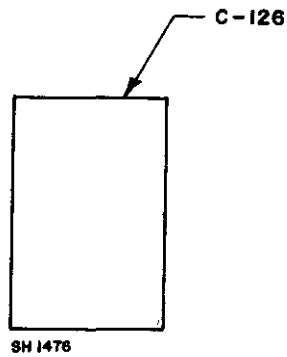


Figure 12 - Description of callouts, etc.

3.9.3 Photographs of the top, bottom and front views of the chassis shall be furnished. When applicable, all the alignment points shall be indicated by arrows (callouts) with appropriate designations (see figures, 13, 14, and 15 which serve as illustrations only).

3.9.3.1 When applicable, an alignment procedure chart shall accompany these drawings and will be included in the same print, if legibility is not sacrificed and overcrowding does not result (see figure 16 which serves as an illustration only).

3.10 Special presentations covered by this specification.-

3.10.1 Where energy flows along a line or channel, sufficient number of arrowheads shall be shown to clearly indicate the direction of flow (see figure 1).

3.10.2 Bridge rectifiers shall be shown schematically.

3.10.3 When possible, mechanically ganged controls shall be connected by dashed lines.

3.10.4 Terminals in a strip shall be drawn in a horizontal or vertical row. An outline of the terminal strip shall be shown to enclose the terminals.

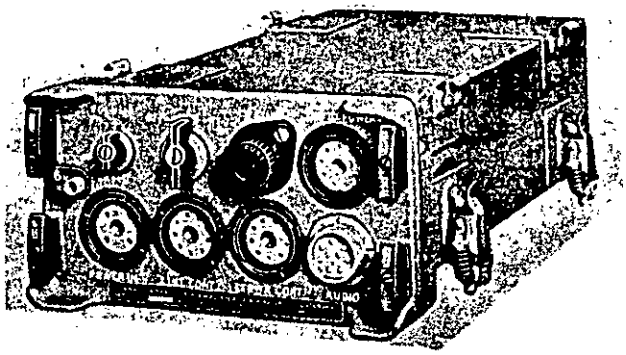
3.10.5 Special test point terminals incorporated in the design of the equipment shall be shown as two small heavily drawn concentric circles (see figure 1), and the abbreviation TP_1, TP_2, shall be used to identify each test point.

3.10.6 Jacks and terminals shall be shown as a small circle drawn lighter than that of the test point circles (see 3.10.5).

3.10.7 The elements of cathode ray display tube which connect to parts shown on the servicing-block diagram shall be depicted schematically in extra heavy lines with the elements abbreviated and numbered. Deflection and focus coils shall be marked as such. An outline of the CRT shall enclose the elements. The words "A" Scope or "PPI" Scope, whichever is applicable, shall be printed near the scope. In addition, the scope tube and symbol number shall be shown inside the scope outline.

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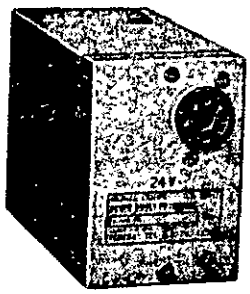
A. F. AMPLIFIER
(Nomenclature)



TECHNICAL CHARACTERISTICS

Maximum power output	1.8 watts
Input impedance	1,500 and 150 ohms
Output impedance	600 ohms
A.f. response	400 - 2500 cycles
Power requirements	Plate 135 volts Fil. 6.3 or 12.6

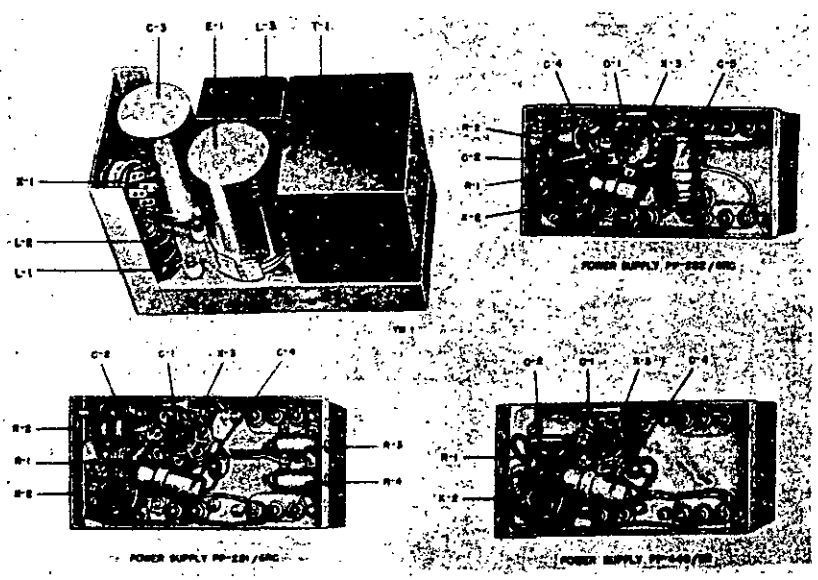
POWER SUPPLY
(Nomenclature)



Output voltage	135 volts 0.118 amp.
Batt. input voltage	6.3 - 12.6 - 25.2 volts d.c.

Figure 13 - Photographic illustrations for representing units.

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NOTES - MODIFICATIONS - CHANGES

Figure 14 - Photographic illustrations for representing units.

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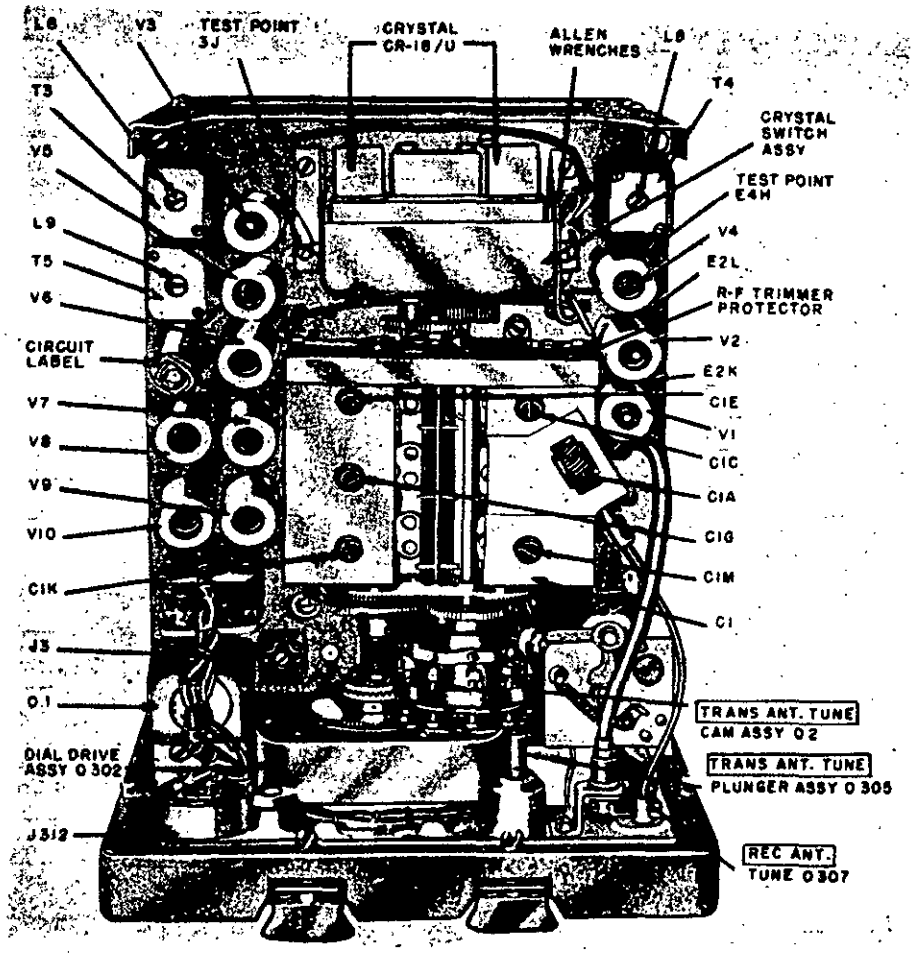


Figure 15 - Photographic illustrations for representing callouts.

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3.9.3.1 When applicable, an alignment procedure chart shall accompany these drawings and will be included in the same print, if legibility is not sacrificed and overcrowding does not result (see figure 16 which serves as an illustration only):

Alignment			
Signal generator	Meter	Adjust	Remarks
4.3 mc. unmod. 0.01 µf to pin 6 V4	Pin 6 V8 and gnd.	Sig. gen.	Test switch to tune Zero beat sig. gen.
4.3 mc. unmod. .01 µf to pin 6 V4	Pin 6 V8 and gnd.	P and S T9	For max-test switch at dial lite off.
4.3 mc. unmod. .01 µf to pin 6 V4	Pin 6 V8 and gnd.	P and S T9	Max. "
4.3 mc. unmod. .01 µf to pin 6 V4	Pin 6 V8 and gnd.	P and S T7	Max. "
4.3 mc. unmod. .01 µf to pin 6 V4	Pin 6 V8 and gnd.	P and S T6	Max. "
4.3 mc. unmod. .01 µf to pin 6 V4	Pin 6 V8 and gnd.	P and S T6	Max. "
4.3 mc. unmod. .01 µf to pin 6 V4	T9 S6 and gnd.	T11	Zero readings between two max. peaks.
Receiver oscillator			
Receiver dial set to highest Cal point. Adjust C39 for zero beat.			
Receiver dial set to lowest Cal point. Adjust T4 for zero beat.			
Receiver dial set to highest Cal point. Adjust C38 for zero beat.			
Receiver r.f.			
55 mc. unmod. 0.01 to Ant.	Pin 6 V8 and gnd.	C3E	Max.
55 mc. unmod. .01 to Ant.	Pin 6 V8 and gnd.	C3C	Max.
55 mc. unmod. .01 to Ant.	Pin 6 V8 and gnd.	C4	Max.
38 mc. unmod. .01 µf to Ant.	Pin 6 V8 and gnd.	T1	Max.
38 mc. unmod. .01 µf to Ant.	Pin 6 V8 and gnd.	T3	Max.
38 mc. unmod. .01 µf to Ant.	Pin 6 V8 and gnd.	T3	Max.

Stage-gain chart

Switch or control	Setting	Switch or control	Setting
SQUELCH TUNE-DIAL LIGHT (ON-OFF) switch (S-3).	OFF, DIAL LIGHT OFF position (except when zero-beating the test signal generator with the built-in tuning oscillator).	VOLUME 6, 12, and 24 VOLTS-OFF-EXTERNAL SUPPLY. CHANNEL dial	Maximum clockwise direction. 6, 12, and 24 VOLTS if vibrator is used. On EXTERNAL SUPPLY if external supply is used. To correspond to the test frequency used. See chart below.

Signal generator at grid of	Approx. input (µv)	Signal generator frequency (mc.)	Limiter meter M-1 reading (volts d.c.)
V-7	87,000	4.3 (G-2)	0.5
V-6	8,700 ± 30%	4.3 (G-2)	.5
V-5	870 ± 45%	4.3 (G-2)	.5
V-4	1,500 ± 75%	4.3 (G-2)	6.5
V-2	500 ± 100% - 50%	4.3 (G-2)	6.5
R-108/GRC (V-1)	30	20.0	6.5
	10	28.0	6.5
R-109/GRC (V-1)	30	28.0	6.5
	10	39.0	6.5
R-110/GRC (V-1)	30	39.0	6.5
	10	55.0	6.5
ANT jack of R-108/GRC	3	20.0	6.5
	1.5	28.0	6.5
ANT jack of R-109/GRC	3	28.0	6.5
	1.5	39.0	6.5
ANT jack of R-110/GRC	3	39.0	6.5
	1.5	55.0	6.5

Figure 16 - Alignment procedure chart.

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3.10.8 As applicable, the transmitter oscillator, duplexer, transmission line (or waveguide), mixer, and local oscillator stages shall be drawn in such a manner as to clearly represent the actual, physical and electrical layout without unduly complicating the diagram.

3.10.9 The following information, as applicable, shall be shown next to the block (or channel) concerned:

(a) Repetition rate.
(b) Receiver intermediate frequency.
(c) Frequency of markers - oscillator and distance represented between markers.
(d) Other frequencies which may be of interest such as "18th Harmonic Phase".

3.10.10 IFF, video synchronizing, and other similar voltages entering or leaving the equipment shown on the servicing-block diagram shall be identified, for example, IFF trigger output and IFF video input.

3.10.11 Operating or adjusting controls associated with any tube block shall be indicated by a dotted line to a small circle, (slotted if screwdriver control) with the circuit designation and function listed (see figure 1). If simplicity is maintained, the actual schematic of the control may be shown.

3.10.12 The d.c. resistance of the primary and secondary windings of all the transformers shall be given. Also, when applicable, the impedance ratio of transformers shall be given (for example, audio output transformer, pulse transformer).

3.10.13 All filament transformers shown on schematic diagrams shall have the voltage and current ratings given for the secondary winding.

3.10.13.1 When waveforms are inverted by the use of a transformer, the polorization of the terminals of the transformer shall be indicated by a dot(.) in the schematic diagrams, and a note shall be made to indicate meaning of symbol. An example is shown on figure 17.

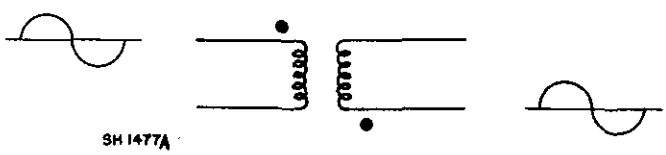


Figure 17 - Respective polarization.

3.10.14 Where damage to the oscilloscope might result, or in certain special instances where the oscilloscope itself may distort or obliterate the waveform, a reference to a note at the bottom of the sheet shall be placed near the waveform concerned. This note shall briefly instruct the technician on the procedure to follow in order to observe the desired waveform.

3.10.15 When terminals, connectors, fuses, switches, and controls appear on the front panel of the equipment these items shall be indicated by an asterisk (*) near the part in reference on the schematic diagrams, and a note shall be placed at the bottom of the print indicating its meaning.

3.10.16 For standardization on symbol designations see ASA Y32.16.

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3.11 Print size, sheet stock, and binding.-

3.11.1 The index and the individual prints listed in 3.1 shall be printed on 9-1/4 by 22 inches sheet stock including the margin dimensions.

3.11.2 Standardization shall be maintained on the following list of presentations:

(a) Heavy lines shall indicate main circuits.
(b) Light lines shall indicate auxiliary circuits or controls.

3.11.3 All blocks and connecting lines shall be drawn in extra heavy lines to furnish the necessary contrast between the general outline of the equipment and miscellaneous information.

3.11.4 The maintenance prints shall be printed on both sides on white opaque rigid vinyl sheets. The base material shall be polyvinyl chloride in accordance with L-P-535 and shall have a minimum thickness of 5 mils. The protective laminate shall be transparent and shall be at least 1 mil thick on each side of the printed sheet in accordance with type G of MIL-I-631. The finished print shall have a matte finish and shall have a thickness no greater than 8 mils. The plastic pages shall be such that they will not be affected by ultra-violet rays when exposed to continuous rays of sunlight.

3.11.5 The maintenance prints shall be printed in black unless use of multicolor is specifically required by the individual equipment specification. In cases where use of multicolor is required, waveforms, fuses, test points, and tube numbers shall be depicted in red.

3.11.5.1 The ink must be fully uniform in coverage and must firmly adhere to the plastic.

3.11.6 Sheets shall be printed on both sides. However, this does not preclude the total number of printed pages ending in an odd number. Prints requiring more space than afforded by a single page 9-1/4 by 22 inches, shall be divided between two facing pages to afford a print size 18-1/2 by 22 inches. For a schematic appearing on two facing pages, the connections between portions of the schematic shall be depicted as vertical lines leaving the bottom side of the top schematic and entering the top side of the lower schematic.

3.11.7 The laminate process shall be such that no visible distortion of the printed materials exists and shall result in the three laminates being completely bonded so there is no visible evidence of an imperfect seal.

3.11.7.1 A margin of no less than 3/8 inch shall be used on the two sides and bottom of the page. For the top margin, see 3.11.8.1.

3.11.8 Unless otherwise specified in the contract or order, the set of maintenance prints shall be bound between two black flexible leatherette covers, or equivalent. The covers shall be approximately 0.018 inch thick and shall be waterproof.

3.11.8.1 Each complete set of maintenance prints shall be bound by noncorrosive type metal, continuous single wire spiral coil through punched holes along the longer dimension of the top in such a manner as to permit the prints to lie flat when opened to any page. The center of the holes shall be approximately 5/16 inch from the top of the page and shall have an additional margin of approximately 1/4 inch down from the hole to the printing.

3.11.8.1.1 The binding shall be of the type which is manufactured under the trade name "Wire-O" by the Wire-O Corporation, Poughkeepsie, N.Y., or its equivalent.

3.11.8.2 The cover shall be titled in heavy large-size type as follows:

MAINTENANCE PRINTS

FOR

RADAR(RADIO) SET NOMENCLATURE

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3.11.9 Security classification.- The security classification of the maintenance prints shall be as designated by the command or agency concerned. The Security Requirements Check List, DD Form 254, which constitutes a part of the contract for all classified material, identifies and indicates the classified features. All pages of classified manuals shall be marked in accordance with Industrial Security Manual for safeguarding classified information.

3.11.9.1 On material that is classified confidential and above, the classification shall be printed in red ink. On unclassified material, the classification shall be printed in black ink.

3.12 Submission of material for approval.-

3.12.1 Two proof copies of the printed material, constituting the complete set of maintenance prints, shall be submitted, together with a sample of laminated material (size 11-1/2 by 22 inches) which complies with 3.11.7 and serves to illustrate the end or final product, to the command or agency concerned for approval prior to publication.

3.12.2 Three copies of the completed set of maintenance prints shall be furnished to the command or agency concerned after publication.

3.13 Index and arrangement of prints.-

3.13.1 An index to the individual prints and tables comprising the set of maintenance prints shall be provided as page one of the set.

3.13.2 Following the index, the tables and prints shall be arranged in the order specified in 3.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Examination.- Prior to shipment, both the manuscript and final maintenance prints shall be examined to determine conformance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Unless otherwise specified in the contract or order, the manufacturer shall enclose two complete sets of maintenance prints with each equipment.

5.2 Each set of maintenance prints shall be enclosed in one or two envelopes depending on the classification. For unclassified sets, a single envelope shall be used; for confidential sets two envelopes shall be used. The outer envelope in either case shall be marked - "This envelope contains material for the maintenance of (nomenclature of equipment)". Only the inner envelope, required for confidential sets, shall be marked "CONFIDENTIAL". For equipment of higher classification, the command or agency concerned shall be consulted.

6. NOTES

6.1 It is the intent that requirements as specified herein shall apply, unless an improved alternate method can be offered and prior approval of the command or agency concerned is received.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 7650-N007SH)

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